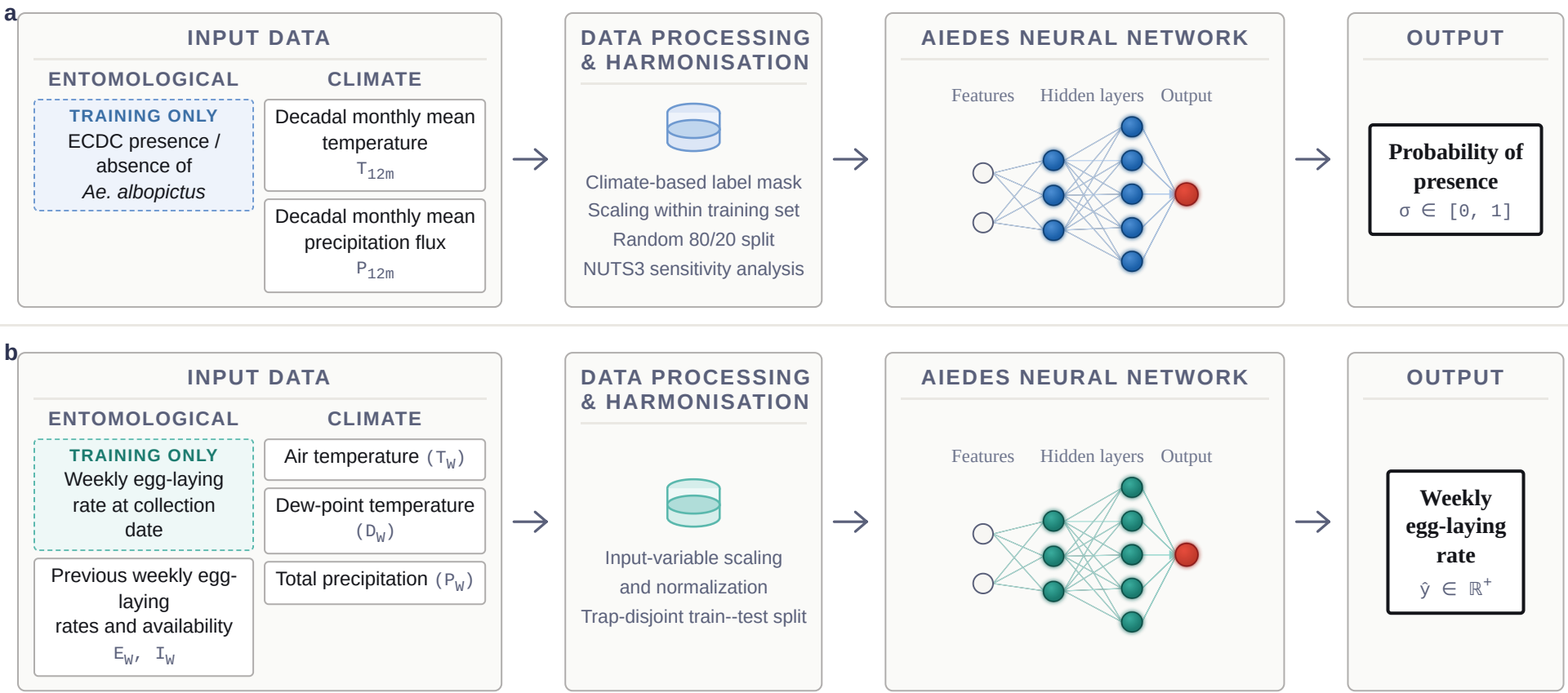

a
INPUT DATA
ENTOMOLOGICAL
TRAINING ONLY
ECDC presence / absence of Ae. albopictus
CLIMATE
Decadal monthly mean temperature
T12m
Decadal monthly mean precipitation flux
P12m
DATA PROCESSING & HARMONISATION
Climate-based label mask
Scaling within training set
Random 80/20 split
NUTS3 sensitivity analysis
AIEDES NEURAL NETWORK
Features
Hidden layers
Output
OUTPUT
Probability of presence
σ ∈ [0, 1]
b
INPUT DATA
ENTOMOLOGICAL
TRAINING ONLY
Weekly egg-laying rate at collection date
Previous weekly egg-laying rates and availability
EW, IW
CLIMATE
Air temperature (TW)
Dew-point temperature (DW)
Total precipitation (PW)
DATA PROCESSING & HARMONISATION
Input-variable scaling and normalization
Trap-disjoint train--test split
AIEDES NEURAL NETWORK
Features
Hidden layers
Output
OUTPUT
Weekly egg-laying rate
ŷ ∈ ℝ+
LEGEND
Covariate (training / evaluation / inference)
Outcome (training / evaluation only) — Aledes -Classifier
Outcome (training / evaluation only) — Aledes -Counter
Hidden unit — Classifier (ReLU / sigmoid)
Hidden unit — Counter (ReLU / softplus)
Output neuron
Feed-forward direction